

# **Week 1**

# **Where Does Data Come From & Tools for Data Engineering**

**ISTA 322 - Data Engineering**

# Recap of last lecture

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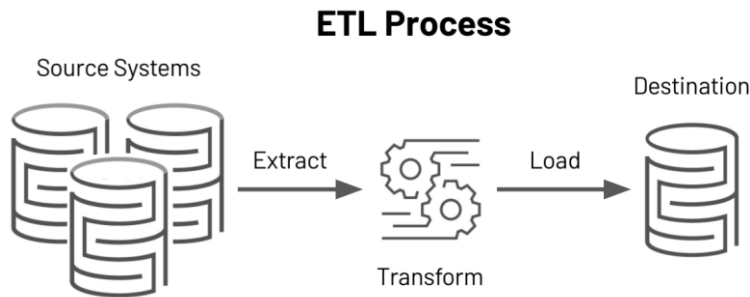
- Data volumes have boomed in the last 20 years
- Some early companies were effective in using this
- This and other things (media, research) drove the potential for data scientists to use big data
- Early efforts did not go well as data is often in messy, not in immediately useful formats
- *And* there was lots of it which limited ability to process
- Early DS roles involved a lot of data engineering
- Now, there are explicit DE roles
- **DE is all about making data useful for analysis**

# Where are we going today?

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- Talk about where all these data are coming from
- The (generally) main job of a DE - making \*ETLs
- Technologies using in DE and what subset we'll use

\*ETL: Extract, transform, load

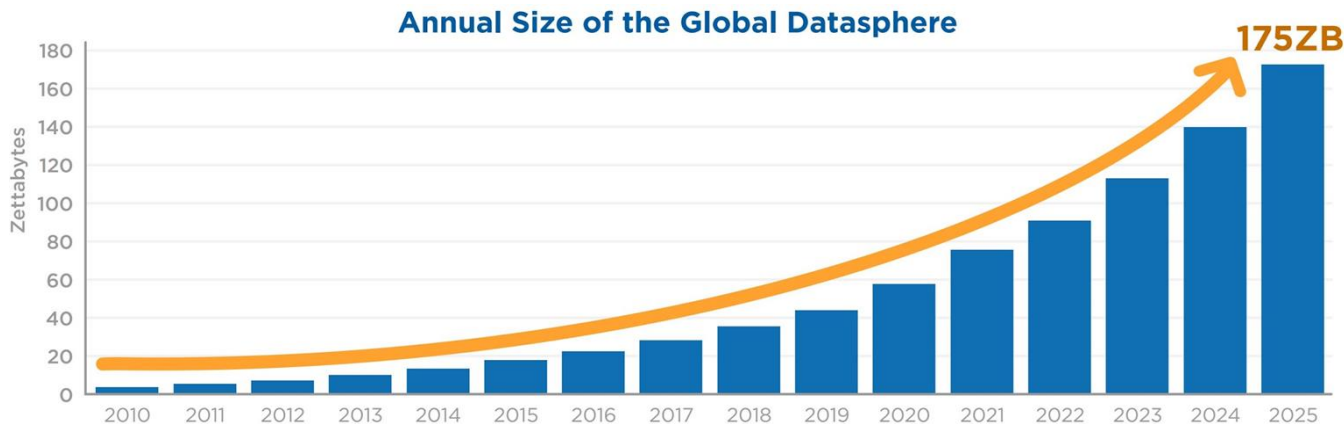


<https://databricks.com/glossary/extract-transform-load>

# Where do data come from?

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- There are tons of data and the amount being collected is exploding.
- What generates these data?
- Events!



**1 Zetabyte is**  
- 350 trillion songs  
- 100k copies of wikipedia

# How events create data

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- Events – data of actions performed by entities
  - Clicked on an ad
  - Left the page
  - Searched for something
  - Tried to log in
  - Made a transaction
  - Scrolled up or down
  - Liked, reacted, retweeted, hearted, shared
  - Uploaded a photo or video
  - Doesn't have to be human... temperature probe recording, machine finishing a job, airplane sensors measuring tons of stuff, etc.
- Also will record time and who did it

# How events create data

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- Events will then be linked to other data that's collected
- e.g. you click on Tiger King
  - {time : 07:26, event : click\_watch, show\_id : tk\_S1E1, user\_id : x88}
- These events are linked to other data that's known about you or the show
- There will be a table that contains show info
  - {show\_id : tk\_S1E1, tags : ['drama', 'reality'], runtime : 55min }
- And user info
  - {user\_id : x88, age : 35, gender : 'M', OS : ['wind', 'andro']}

# Event data at Netflix

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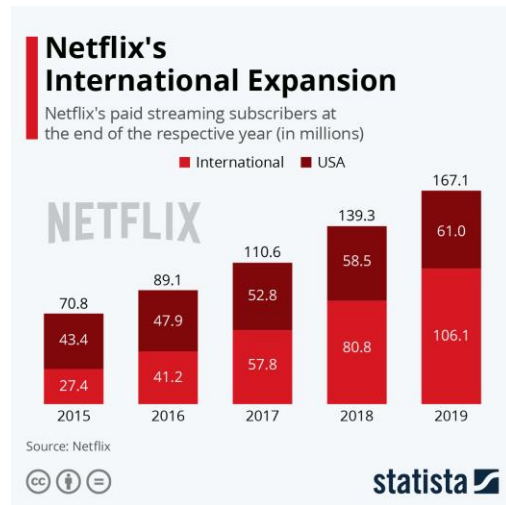


- With this data arriving at over **2 million events per second**, getting it into a database that can be queried quickly is formidable. We need sufficient dimensionality for the data to be useful in isolating issues and as such **we generate over 115 billion rows per day**.
  - This was in 2020 - [Ref @ Netflix Tech Blog](#)
- They ‘only’ generated 10 billion rows a day in 2015
  - [Ref @ Netflix Tech Blog](#)

# How events create data

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- In just 5 years Netflix increased the amount of data collected by 10x (115 billion vs 10 billion)
  - Number of subscribers only increased by 2.5 - [ref](#)
  - Increased granularity of data collected
- This allows for more complex models & better analytics
  - “We need sufficient dimensionality for the data to be useful in isolating issues”
  - Remember, models need  $n \times m$  matrix
  - More dimensions = more features in the matrix
  - More features = more models & better predictions
  - $\therefore$  more money





# Not all events

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- Of course, not all data collected is structured like this
- Some is just stored in a database across multiple tables
  - Each transaction in a convenience store

| TABLE ID: STORE |             |         |
|-----------------|-------------|---------|
| store_id        | store_state | country |
| az_23           | AZ          | USA     |
| az_45           | AZ          | USA     |
| ca_12           | CA          | USA     |
| to_39           | Ontario     | Canada  |

| TABLE ID: TRANSACTIONS |          |       |       |
|------------------------|----------|-------|-------|
| transact_id            | store_id | UPC   | price |
| x88943                 | az_23    | 49914 | 2.57  |
| x88943                 | az_23    | 99371 | 1.99  |
| a85921                 | to_39    | 95831 | 8.99  |
| a85921                 | to_39    | 99492 | 5.49  |
| a85921                 | to_39    | 27482 | 4.49  |
| z88930                 | az_45    | 33491 | 0.99  |

# Not all events

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- Of course, not all data collected is structured like this
- Some is just stored in a database across multiple tables
  - Each transaction in a convenience store
  - And data collected might not be optimized

| A    | B                 | C       | D               | E               | F             | G        | H         | I               | J     |
|------|-------------------|---------|-----------------|-----------------|---------------|----------|-----------|-----------------|-------|
| id   | name              | host_id | host_name       | neighbourhood_c | neighbourhood | latitude | longitude | room_type       | price |
| 2539 | Clean & quiet apt | 2787    | John            | Brooklyn        | Kensington    | 40.64749 | -73.97237 | Private room    | 149   |
| 2595 | Skylit Midtown C  | 2845    | Jennifer        | Manhattan       | Midtown       | 40.75362 | -73.98377 | Entire home/apt | 225   |
| 3647 | THE VILLAGE O     | 4632    | Elisabeth       | Manhattan       | Harlem        | 40.80902 | -73.9419  | Private room    | 150   |
| 3831 | Cozy Entire Floor | 4869    | LisaRoxanne     | Brooklyn        | Clinton Hill  | 40.68514 | -73.95976 | Entire home/apt | 89    |
| 5022 | Entire Apt: Spaci | 7192    | Laura           | Manhattan       | East Harlem   | 40.79851 | -73.94399 | Entire home/apt | 80    |
| 7322 | Chelsea Perfect   | 18946   | Doti            | Manhattan       | Chelsea       | 40.74192 | -73.99501 | Private room    | 140   |
| 7726 | Hip Historic Brow | 20950   | Adam And Charit | Brooklyn        | Crown Heights | 40.67592 | -73.94694 | Entire home/apt | 99    |
| 7750 | Huge 2 BR Uppe    | 17985   | Sing            | Manhattan       | East Harlem   | 40.79685 | -73.94872 | Entire home/apt | 190   |
| 7801 | Sweet and Spaci   | 21207   | Chaya           | Brooklyn        | Williamsburg  | 40.71842 | -73.95718 | Entire home/apt | 299   |
| 8024 | CBG CtyBGd He     | 22486   | Lisel           | Brooklyn        | Park Slope    | 40.68069 | -73.97706 | Private room    | 130   |
| 8025 | CBG Helps Haiti   | 22486   | Lisel           | Brooklyn        | Park Slope    | 40.67989 | -73.97798 | Private room    | 80    |
| 8110 | CBG Helps Haiti   | 22486   | Lisel           | Brooklyn        | Park Slope    | 40.68001 | -73.97865 | Private room    | 110   |

# So what does a DE do again?

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- Takes data from these various databases that are recording events/transactions/information
- Reorganizes it in some way or another into a format that lets people do analytics or data science
- Puts it in a database for them to use.
- This process has a general name - **ETL**
  - **Extract - Transform - Load**

# ETL

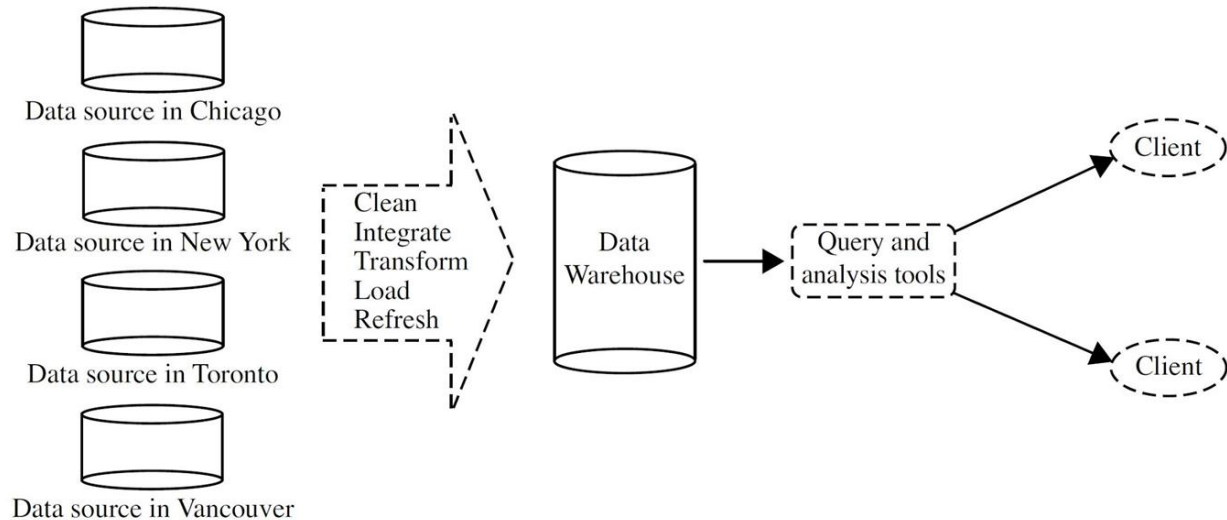
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- **ETLs are essentially the core of DE**
- That raw data in structured, semi-structured, or unstructured format is all stored in a **data lake**
- The transform step is going to remove errors, create features, scale values, aggregate data for metrics and whatever else is needed to support analytics and DS
- The transformed data is stored in a **data warehouse**

# ETL

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- [From reading - Ch1 Data Mining Concepts and Techniques](#)



**Figure 1.6** Typical framework of a data warehouse for *AllElectronics*.

# But how to deal with so many events?

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- OK, our goal is to get the data into a useful format
- But we're dealing **lots** of data
- Average computer has say 16gb of memory
  - A decade ago Facebook was dealing with 10+ gb of processed data a day
  - Amazon's daily login datafile alone is 1tb
- Obviously this is the other challenge of DE
  - How to deal with massive volumes of data fast enough to be useful
  - Can't let it take hours/days/weeks to process on one machine

# Enter big data technologies

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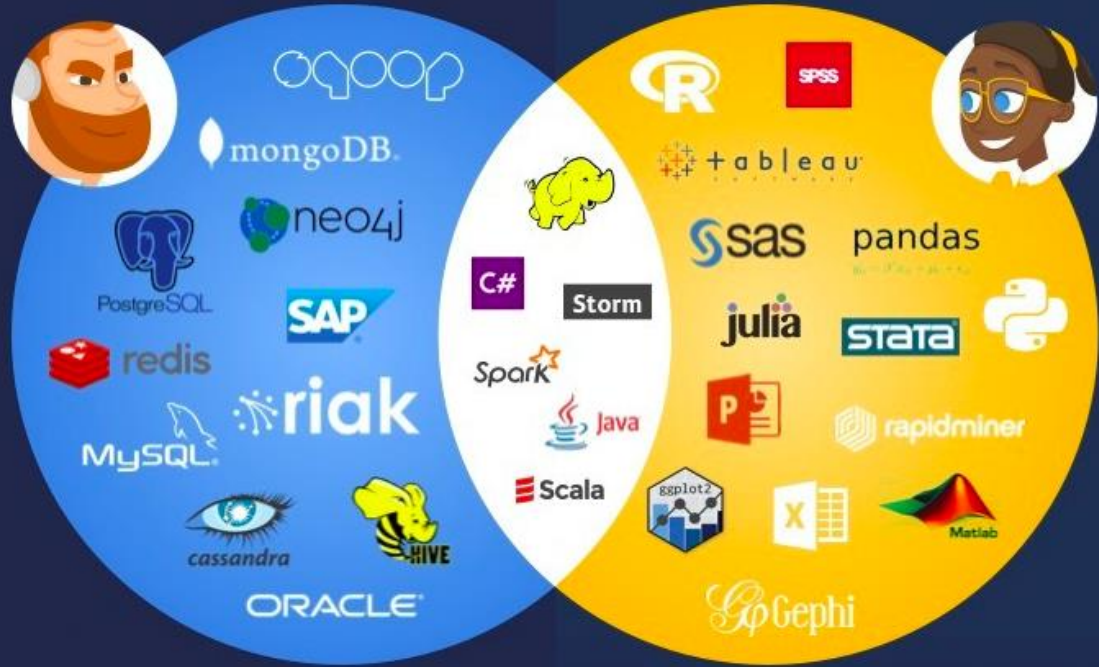
- The other part of being a DE is using big data processing frameworks that allow for much, much faster data processing
- Technologies like hadoop/mapreduce and Spark utilized clusters of machines to distribute framework and optimize speed

# Enter big data tech

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- The other part processing from data processing
- Technologies to clusters of machines speed

## Languages, Tools & Software





# Enter big data technologies

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- The other part of being a DE is using big data processing frameworks that allow for much, much faster data processing
- Technologies like hadoop/mapreduce and Spark utilized clusters of machines to distribute framework and optimize speed
- It's a massive ecosystem of tools - We're only going to learn some of the essential tools

# A bit more about the technologies we're going to use

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- Languages / technologies

- Python and pandas
- SQL - Likely PostgreSQL
- Pyspark locally
- Pyspark via Databricks

- Environments

- We'll be working in Jupyter Notebooks
- Use [Google Colaboratory](#) - Google cloud based Jupyter Notebook
  - You'll download a notebook, upload and open there
- You're welcome to use a local install, but I won't be providing tutorials (I can't troubleshoot 40 installs of all the libraries)
- AWS - Pull from and set up database on AWS
- [Databricks](#) - Cloud notebook based analytics/DS platform